

# THE MOD-2 COKERNEL DENSITY OF A HECKE OPERATOR ON A NON-ANALYTIC PRO-2 TOWER: SPLIT FORM, EXPLICIT KERNELS, AND BOUNDS

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ABSTRACT. Let  $P = (1 + \varphi\mathcal{O})/(1 + t\mathbb{F}_2[[t]])$  be the pro-2 group attached by R. Chen to the Cartwright–Mantero–Steger–Zappa  $\tilde{A}_2$ -lattice of order  $q = 2$ , where  $\mathcal{O} = \mathbb{F}_8[[\varphi; \sigma]]$ ,  $\varphi^3 = t$ , and let  $F \in \mathbb{F}_2[\Gamma^0]$  be the element whose right-multiplication cokernel computes the  $\mathbb{F}_2$ -corank of the Hecke Laplacian along the canonical tower. Chen asked whether the cokernel density  $c_e$  vanishes, and whether the restricted enveloping algebra of  $\mathrm{gr} P$  is a domain. We answer the second question negatively, exhibiting an explicit square-zero element in degree 4. We show that  $\mathrm{gr} P \otimes \mathbb{F}_8$  is the positive part of the affine Lie algebra  $A_2^{(1)}$  in its principal grading, so that  $\mathrm{gr} P$  is a twisted  $\mathbb{F}_2$ -form of it, and that the leading form of  $F$  is homogeneous of weight equal to the null root  $\delta$ . This yields explicit kernel elements for every level (governed by a “binary odometer” along the imaginary root chain  $3 \cdot 2^r$ ), monotonicity statements, and, by exact computation up to  $|G_{10}| = 2^{24}$ , the bound  $c_e \leq 0.030322$  (previously 0.0645). Using evaluation representations of the loop algebra we prove that the leading form of  $F$  is a non-zero-divisor in  $\mathrm{gr} \mathbb{F}_2[[P]]$ , hence that  $F$  is a non-zero-divisor in  $\mathbb{F}_2[[P]]$ . Finally we prove that  $c_e = 0$  follows from two explicit hypotheses on the kernel of the leading form (verified for all levels  $m \leq 10$ ), and we give computational evidence that the zero divisors of  $\mathrm{gr} \mathbb{F}_2[[P]]$  do not lift to  $\mathbb{F}_2[[P]]$ , consistent with Jaikin-Zapirain’s pro- $p$  Atiyah conjecture.

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## 1. INTRODUCTION

**1.1. The question.** Let  $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$  with  $\alpha^3 = \alpha + 1$ ,  $\sigma(a) = a^2$ , and let  $D$  be the cyclic division algebra  $(\mathbb{F}_8(t)/\mathbb{F}_2(t), \sigma, t)$ , i.e.  $D = \bigoplus_{j=0}^2 \mathbb{F}_8(t)\varphi^j$  with  $\varphi a = \sigma(a)\varphi$  and  $\varphi^3 = t$ . Chen [1, Thm. 6.4.1] shows that  $a_x \mapsto 1 + \alpha^x \varphi$  ( $x \in \mathbb{Z}/7$ ) embeds the Cartwright–Mantero–Steger–Zappa triangle group  $\Gamma$  of order  $q = 2$  [2] into  $D^\times/\mathbb{F}_2(t)^\times$ , so that  $\Gamma$  is a lattice in  $PGL_3(\mathbb{F}_2((s)))$ ,  $s = 1 + t$ . Completing at  $t = 0$ , with maximal order  $\mathcal{O} = \mathbb{F}_8[[\varphi; \sigma]]$ , one obtains the pro-2 group

$$P = (1 + \varphi\mathcal{O})/(1 + t\mathbb{F}_2[[t]]), \quad P_m = \text{im}(1 + \varphi^m\mathcal{O}), \quad G_m = P/P_m,$$

with  $|G_m| = 2^{N_m}$ ,  $N_m = 3(m-1) - \lfloor (m-1)/3 \rfloor$ . The type-preserving subgroup  $\Gamma^0$  is dense in  $P$ ;  $P$  is torsion-free,  $d(P) = 3$ , of infinite rank and not  $p$ -adic analytic [1, Thm. 6.4.2].

With Schreier generators  $\theta_{s,x}$  and  $\bar{f}_s = \sum_{x \in \mathbb{Z}/7} \theta_{s,x}^{-1} \in \mathbb{F}_2[G_m]$  put

$$F = 1 + \bar{f}_1 \bar{f}_0 \bar{f}_2, \quad C_m = \mathbb{F}_2[G_m]/\mathbb{F}_2[G_m]F, \quad n_m = 3|G_m|.$$

Then  $\dim C_m - 1$  is the number of even invariant factors of the Hecke Laplacian of the level- $m$  quotient complex [1, Prop. 6.8.3],  $\dim C_m/n_m$  is non-increasing, and

$$c_e := \lim_{m \rightarrow \infty} \dim C_m/n_m$$

controls the linear growth of the 2-part of the Hecke Jacobian:  $\liminf v_2|\text{Jac}^H(Y_m)|/n_m \geq c_e$  [1, Thm. 6.8.8]. Chen asks whether  $c_e = 0$ , and (Remark 6.9.2 of [1]) whether the restricted enveloping algebra of  $\text{gr } P$  is a domain, which would make  $\mathbb{F}_2[[P]]$  a domain.

**1.2. Relation to the pro- $p$  Atiyah conjecture.** By Jaikin-Zapirain [3, Prop. 11.2], for any pro- $p$  group  $G$  and any chain of open normal subgroups with trivial intersection,  $\text{rk}_G(x) = \lim \text{rk}_{\mathbb{F}_p}(x \text{ on } \mathbb{F}_p[G/U_i])/|G : U_i|$  exists and does not depend on the chain. Hence

$$c_e = \frac{1}{3}(1 - \text{rk}_P(F))$$

is intrinsic to  $(P, F)$ . The pro- $p$  Atiyah conjecture [3, Conj. 11.1] for the torsion-free group  $P$  would give  $\text{rk}_P(F) = 1$ , i.e.  $c_e = 0$ . It is known for residually torsion-free  $p$ -adic analytic groups, free pro- $p$  groups and virtually free-by- $\mathbb{Z}_p$  groups [4], none of which covers  $P$ .

**1.3. Main results.** Write  $L = \text{gr } P = \bigoplus_{k \geq 1} L_k$  for the graded restricted Lie algebra of the filtration  $(P_k)$ ,  $u(L)$  for its restricted enveloping algebra and  $u_m = u(L/L_{\geq m}) \cong \text{gr } \mathbb{F}_2[G_m]$  (Jennings–Quillen). Let  $\sigma F \in u(L)_3$  be the leading form of  $F$ .

**Theorem A** (Section 3).  *$u(L)$  is not a domain. There is a 3-dimensional subspace  $V \subset u(L)_4$  with  $V \cdot V = 0$ ; in particular every  $0 \neq v \in V$  satisfies  $v^2 = 0$ . There are no homogeneous zero-divisor pairs of total degree  $\leq 7$ . Hence the answer to Remark 6.9.2 of [1] is negative.*

**Theorem B** (Section 4).  *$L \otimes_{\mathbb{F}_2} \mathbb{F}_8$  is isomorphic, as a graded restricted Lie algebra, to the positive part  $\widehat{\mathfrak{n}}$  of the affine Lie algebra of type  $A_2^{(1)}$  in its principal grading, with 2-map given by matrix squaring.  $L$  is the  $\mathbb{F}_2$ -form of  $\widehat{\mathfrak{n}}$  twisted by Frobenius composed with the rotation of the affine Dynkin diagram. Under this isomorphism the leading form  $\sigma F$  becomes*

$$\sigma' = e_0 e_1 e_2 + e_2[e_0, e_1] + e_0[e_1, e_2] + h, \quad h \in \widehat{\mathfrak{n}}_\delta,$$

which is homogeneous of weight  $\delta = \alpha_0 + \alpha_1 + \alpha_2$  for the affine root grading.

**Theorem C** (Sections 5–6). *For every  $m \geq 2$  there are three explicit monomials  $\omega_0, \omega_1, \omega_2 \in u_m$  of degree*

$$d(m) = \sum_{k \in \mathcal{H}_m} k, \quad \mathcal{H}_m = \{1 \leq k < m : 3 \nmid k\} \cup \{3 \cdot 2^r < m\},$$

such that  $u_m \omega_0 + u_m \omega_1 + u_m \omega_2$  lies in the kernel of right multiplication by  $\sigma F$ . Moreover the first non-injective degree  $k_m$  of  $R_{\sigma F}$  is non-decreasing in  $m$ , the kernel density  $\dim \ker R_{\sigma F} / |G_m|$  is non-increasing in  $m$ , and  $k_m \leq d(m) \sim m^2/3$ .

**Theorem D** (Section 8; exact computation).  $\dim C_7 = 10226$ , and the graded cokernel of  $R_{\sigma F}$  on  $u_m$  has dimension 64216, 423412, 1526176 for  $m = 8, 9, 10$ . Consequently

$$c_e \leq \frac{1526176}{3 \cdot 2^{24}} = 0.030322 \dots$$

In Section 7 we prove that right and left multiplication by the leading form are injective on the full algebra  $u(\hat{\mathfrak{n}})$  (Theorem 7.1); thus  $F$  is a non-zero-divisor in  $\mathbb{F}_2[[P]]$  and  $k_m \geq m - 3$ . In Section 9 we split the kernel density as a maximal-rank baseline, which we show is  $O(m^{-1/2})$  (numerically  $\approx 2.55 m^{-3/2}$ ), plus an excess  $e_m$  of order  $10^{-3}$ , and formulate  $c_e = 0$  as a Lefschetz-type statement (Conjecture 9.2). We prove that  $c_e = 0$  follows from a growth bound on the kernel below the middle degree together with  $k_m \geq d(m) - 1$  (Theorem 9.4); both hypotheses hold for  $m \leq 10$ , and we show that the excess of  $\sigma'$  is a special-position effect: a generic element of the same weight has maximal rank on every block for  $m \leq 8$ . In Section 10 we show by exhaustive linear algebra that every zero-divisor pair of Theorem A is obstructed at the first or second lifting step in  $\mathbb{F}_2[[P]]$ ; thus they are artefacts of the filtration and give no counterexample to the pro-2 Kaplansky conjecture.

**Status of the statements.** Results labelled *Theorem*, *Proposition* or *Lemma* have complete proofs here, possibly relying on a finite exact computation which is then named explicitly (“proof by computation”). All computations are exact linear algebra over  $\mathbb{F}_2$  and were cross-checked by independent implementations; see Appendix A.

## 2. THE GRADED ALGEBRA

**2.1. The Lie algebra  $L$ .** Let  $A = \mathbb{F}_8[\varphi; \sigma]$  (graded by  $\deg \varphi = 1$ ); its centre is  $\mathbb{F}_2[t]$ ,  $t = \varphi^3$ .

**Lemma 2.1.** *The filtration  $(P_k)$  is a restricted  $N$ -series and*

$$L \cong A_{>0}/T, \quad T = t\mathbb{F}_2[t],$$

as graded restricted Lie algebras, where  $A_{>0}$  carries the commutator bracket and the squaring map. Explicitly,  $L_k = \mathbb{F}_8 \varphi^k$  for  $3 \nmid k$  and  $L_k = \mathbb{F}_8 \varphi^k / \mathbb{F}_2 t^{k/3}$  for  $3 \mid k$ , and

$$[a\varphi^i, b\varphi^j] = (a\sigma^i(b) + b\sigma^j(a))\varphi^{i+j}, \quad (a\varphi^k)^{[2]} = a\sigma^k(a)\varphi^{2k}.$$

*Proof.* For  $x = 1 + a\varphi^i + \dots$ ,  $y = 1 + b\varphi^j + \dots$  one has  $[x, y] \equiv 1 + (a\sigma^i(b) + b\sigma^j(a))\varphi^{i+j}$  and  $x^2 \equiv 1 + a\sigma^i(a)\varphi^{2i}$  modulo higher order; these are the commutator and square in  $A$ . The centre  $T$  is closed under squaring, hence a restricted ideal, and quotienting by  $1 + t\mathbb{F}_2[[t]]$  removes exactly it.  $\square$

Write  $d_k = \dim_{\mathbb{F}_2} L_k$ , so  $d_k = 3$  for  $3 \nmid k$  and  $d_k = 2$  for  $3 \mid k$ . In particular  $d_{2k} = d_k$ .

**Proposition 2.2.** *The Hilbert series of  $u(L)$  is*

$$\sum_j \dim u(L)_j z^j = \prod_{k \geq 1} (1 + z^k)^{d_k} = \prod_{k \text{ odd}} (1 - z^k)^{-d_k}.$$

*In particular  $u(L)$  has the Hilbert series of a commutative polynomial ring with  $d_k$  generators in each odd degree  $k$ , and has subexponential growth.*

*Proof.* The first equality is PBW for restricted enveloping algebras. For the second use  $d_{2^j k} = d_k$  and  $\prod_{j \geq 0} (1 + x^{2^j}) = (1 - x)^{-1}$ .  $\square$

**2.2. The leading form of  $F$ .** By Jennings–Quillen [5, 6],  $\text{gr}_I \mathbb{F}_2[G_m] \cong u(L/L_{\geq m}) =: u_m$ , where  $I$  is the augmentation ideal; in degrees  $< m$ ,  $u_m$  agrees with  $u(L)$ .

**Proposition 2.3** (proof by computation).  *$F \in I^3 \setminus I^4$ , and its class  $\sigma F \in u(L)_3$  is, independently of  $m \geq 4$ ,*

$$\sigma F = x_{1,1}x_{1,\alpha}x_{1,\alpha^2} + x_{1,1}x_{2,\alpha} + x_{1,\alpha^2}x_{2,\alpha} + x_{1,1}x_{2,\alpha^2} + x_{1,\alpha}x_{2,\alpha^2},$$

*where  $x_{k,b}$  is the class of  $1 + b\varphi^k$  and PBW monomials are ordered by  $(k, b)$ .*

**Lemma 2.4** (semicontinuity). *For every  $m$ ,  $\dim C_m \leq \dim \text{coker}(R_{\sigma F} : u_m \rightarrow u_m)$ , where  $R_{\sigma F}(y) = y\sigma F$ . Consequently  $c_e \leq \dim \text{coker}(R_{\sigma F}|u_m)/n_m$  for every  $m$ .*

*Proof.* Right multiplication by  $F$  maps  $I^j$  into  $I^{j+3}$  and induces  $R_{\sigma F}$  on the associated graded. For a filtered map  $f$  of finite-dimensional filtered spaces,  $\text{rk } f \geq \text{rk gr } f$ . The second claim follows from the monotonicity of  $\dim C_m/n_m$  [1, Prop. 6.8.3].  $\square$

### 3. $u(L)$ IS NOT A DOMAIN

**Theorem 3.1** (proof by computation). *Let*

$$\begin{aligned} v = & x_{1,1}x_{1,\alpha}x_{2,1} + x_{1,\alpha}x_{1,\alpha^2}x_{2,1} + x_{1,\alpha}x_{1,\alpha^2}x_{2,\alpha} + x_{1,1}x_{1,\alpha}x_{2,\alpha^2} \\ & + x_{1,1}x_{1,\alpha^2}x_{2,\alpha^2} + x_{1,\alpha}x_{1,\alpha^2}x_{2,\alpha^2} + x_{2,1}x_{2,\alpha^2} + x_{2,\alpha}x_{2,\alpha^2} \in u(L)_4. \end{aligned}$$

*Then  $v \neq 0$  and  $v^2 = 0$  in  $u(L)_8$ . More precisely, there is a 3-dimensional subspace  $V \ni v$  of  $u(L)_4$  with  $V \cdot V = 0$ , and the left and right kernels of multiplication by  $v$  on  $u(L)_4$  are both equal to  $V$ . Every homogeneous pair  $x, y \neq 0$  with  $xy = 0$  has  $\deg x + \deg y \geq 8$ .*

*Verification.* (i) In a PBW implementation of  $u(L)$  (Lemma 2.1), checked against the Hilbert series, associativity on random triples, and the group algebra (Jennings products in  $\mathbb{F}_2[G_7]$  and  $\mathbb{F}_2[G_{12}]$ ),  $v^2 = 0$  and  $V \cdot V = 0$ . (ii) Independently of that code: lifting  $v$  to an element  $\tilde{v} \in \mathbb{F}_2[G_9]$  supported on 18 group elements, the product  $\tilde{v}\tilde{w}$  ( $w \in V$ ) has no Jennings coordinates of weight  $\leq 8$ , so  $\tilde{v}\tilde{w} \in I^9$ , which by Jennings' basis theorem is the statement  $vw = 0$  in  $\text{gr}_8 \mathbb{F}_2[G_9] = u(L)_8$ . (iii) The last assertion is an exhaustive rank computation: for all nonzero homogeneous  $x$  of degree  $a \leq 3$  and the stated ranges of  $b$  (degree 1:  $b \leq 23$ ; degree 2:  $b \leq 20$ ; degree 3:  $b \leq 13$ ; degree 4:  $b \leq 3$ ) left and right multiplication  $u_b \rightarrow u_{a+b}$  is injective.  $\square$

**Remark 3.2.** A second family exists in degrees  $(3, 14)$ :  $x = x_{1,1}x_{2,1} + x_{1,\alpha}x_{2,\alpha} + x_{1,\alpha}x_{2,\alpha^2} + x_{1,\alpha^2}x_{2,\alpha^2}$  annihilates a 3-dimensional subspace of  $u(L)_{14}$ . By contrast,  $\sigma F$  is not a zero divisor in low degrees: left and right multiplication  $u_b \rightarrow u_{b+3}$  by  $\sigma F$  is injective for all  $b \leq 28$ . The conceptual source of these zero divisors is explained by Theorem B: after  $\otimes \mathbb{F}_8$  the real root vectors square to zero, and  $L$  only hides this through its twisted  $\mathbb{F}_2$ -structure.

## 4. THE SPLIT FORM

Let  $E = \mathbb{F}_2^3$  with idempotents  $e_0, e_1, e_2$ ,  $s(e_p) = e_{p+1}$  (indices mod 3),  $A^s = E[\varphi; s]$ ,  $T^s = t\mathbb{F}_2[t]$  with  $t = (1, 1, 1)\varphi^3$ , and  $L^s = A_{>0}^s/T^s$ .

**Proposition 4.1.** *The map  $\psi : A \otimes_{\mathbb{F}_2} \mathbb{F}_8 \rightarrow A^s \otimes_{\mathbb{F}_2} \mathbb{F}_8$ ,  $a\varphi^k \otimes \lambda \mapsto (\lambda a, \lambda\sigma^{-1}(a), \lambda\sigma^{-2}(a))\varphi^k$ , is an isomorphism of graded  $\mathbb{F}_8$ -algebras with  $\psi(T \otimes \mathbb{F}_8) = T^s \otimes \mathbb{F}_8$ . Hence  $L \otimes \mathbb{F}_8 \cong L^s \otimes \mathbb{F}_8$ ,  $u(L) \otimes \mathbb{F}_8 \cong u(L^s) \otimes \mathbb{F}_8$  and  $u_m \otimes \mathbb{F}_8 \cong u_m^s \otimes \mathbb{F}_8$ .*

*Proof.* In degree 0 this is the classical isomorphism  $\mathbb{F}_8 \otimes \mathbb{F}_8 \cong \prod_{\text{Gal}} \mathbb{F}_8$ , which intertwines  $\sigma \otimes 1$  with the shift  $s$ . It therefore extends to the skew polynomial rings with  $\varphi \mapsto \varphi$ , bijectively in each degree, and  $t \mapsto (1, 1, 1)\varphi^3$ . Restricted structures are defined by products.  $\square$

**Proposition 4.2.** *Let  $\Pi = E_{10} + E_{21} + tE_{02} \in M_3(\mathbb{F}_2[t])$ . Then  $\sum_k D_k \varphi^k \mapsto \sum_k D_k \Pi^k$  ( $D_k$  diagonal) identifies  $A^s$  with the Iwahori order  $\{X \in M_3(\mathbb{F}_2[t]) : X(0) \text{ lower triangular}\}$ , graded by the principal grading  $\deg E_{ij}t^n = 3n + (i - j)$ , and identifies  $L^s$  with the nilradical of the standard Iwahori subalgebra of  $\mathfrak{sl}_3 \otimes \mathbb{F}_2[t, t^{-1}]$ , i.e. with the positive part  $\hat{\mathfrak{n}}$  of  $A_2^{(1)}$  in the principal grading. The degree-one elements  $e_p\varphi$  are the Chevalley generators; real root vectors square to zero and  $(ht^n)^{[2]} = h^2t^{2n}$  for Cartan elements.*

*Proof.*  $\Pi E_{qq} = E_{q+1, q+1}\Pi$  and  $\Pi^3 = tI$ , so the map is an injective graded algebra homomorphism with the stated image. Since 3 is invertible in  $\mathbb{F}_2$ ,  $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus \mathbb{F}_2 I$ , and  $x \mapsto x + \text{tr}(x)I$  identifies  $\mathfrak{gl}_3/\mathbb{F}_2 I$  with  $\mathfrak{sl}_3$  as restricted Lie algebras ( $\text{tr}(x^2) = \text{tr}(x)^2$  in characteristic 2). The dimension count  $d_k = 3$  ( $3 \nmid k$ : three real roots of principal height  $k$ ) and  $d_k = 2$  ( $3 \mid k$ : imaginary root of multiplicity  $\text{rank } \mathfrak{sl}_3$ ) is consistent.  $\square$

**4.1. Gradings and symmetries.**  $A^s$  is the path algebra of the oriented 3-cycle, so  $L^s$ ,  $u(L^s)$  and  $u_m^s$  are graded by  $Q_+ = \mathbb{N}\beta_0 \oplus \mathbb{N}\beta_1 \oplus \mathbb{N}\beta_2$  ( $\beta_p$  the weight of  $e_p\varphi$ , the simple roots), refining the principal degree;  $t$  has weight  $\delta = \beta_0 + \beta_1 + \beta_2$ . The rotation  $\rho(e_p\varphi^k) = e_{p+1}\varphi^k$  is an automorphism, and  $\tau(e_p\varphi^k) = e_{k-p}\varphi^k$  is an anti-automorphism of  $A^s$  fixing  $T^s$ ; in characteristic 2 it induces a restricted Lie automorphism of  $L^s$  and an anti-automorphism  $\Theta'$  of  $u_m^s$ .

**Proposition 4.3** (proof by computation for the first identity). *With  $y_{k,p} = e_p\varphi^k$ ,*

$$\sigma' := \psi(\sigma F) = y_{1,0}y_{1,1}y_{1,2} + y_{1,2}y_{2,1} + y_{1,0}y_{2,2} + y_{3,1} = c_1(y_{1,1}y_{1,2} + y_{2,2}) + c_2 y_{1,2} + c_3,$$

*where  $c_k := e_{k-1}\varphi^k$ . In particular  $\sigma'$  has  $\mathbb{F}_2$ -coefficients, is  $Q$ -homogeneous of weight  $\delta$ , and  $\rho(\sigma') = \Theta'(\sigma') = \sigma'$ .*

**Corollary 4.4.** *For all  $m, n$ ,  $\text{rk}(R_{\sigma F}|(u_m)_n) = \text{rk}(R_{\sigma'}|(u_m^s)_n)$ , and  $R_{\sigma'} = \bigoplus_{\gamma} R_{\gamma}$  with  $R_{\gamma} : (u_m^s)_{\gamma} \rightarrow (u_m^s)_{\gamma+\delta}$ . Moreover  $\text{rk } R_{\gamma} = \text{rk } R_{\rho\gamma}$ , and  $\text{rk } R_{\gamma} = \text{rk } R_{\theta(\Gamma-\gamma-\delta)}$ , where  $\Gamma$  is the weight of the top degree of  $u_m^s$  and  $\theta(n_0, n_1, n_2) = (n_1, n_0, n_2)$ .*

*Proof.* Ranks are invariant under extension of scalars,  $\psi$  carries  $\sigma F$  to  $\sigma'$ , and  $\sigma'$  is defined over  $\mathbb{F}_2$ ;  $\sigma'$  has weight  $\delta$ , and  $\rho$  is an automorphism fixing  $\sigma'$  and mapping block  $\gamma$  to  $\rho\gamma$ . For the duality,  $u_m^s$  is a finite-dimensional connected graded Hopf algebra whose top degree is one-dimensional (spanned by the product of all PBW letters, a two-sided integral). Let  $\lambda$  be the projection onto the top degree. The pairing  $(a, b) \mapsto \lambda(ab)$  is nondegenerate on  $(u_m^s)_{\gamma} \times (u_m^s)_{\Gamma-\gamma}$ :  $u_m^s$  is Frobenius [7] and, for graded connected algebras, any Frobenius form may be taken to be this projection. Since  $\lambda((y\sigma')z) = \lambda(y(\sigma'z))$ , the map  $R_{\gamma}$  is adjoint to left multiplication  $L_{\sigma'} : (u_m^s)_{\Gamma-\gamma-\delta} \rightarrow (u_m^s)_{\Gamma-\gamma}$ . Finally  $\Theta'$  is an anti-automorphism with  $\Theta'(\sigma'z) = \Theta'(z)\sigma'$ , mapping weight  $\mu$  to  $\theta(\mu)$ , so it conjugates  $L_{\sigma'}$  on block  $\mu$  to  $R_{\sigma'}$  on block  $\theta(\mu)$ . (The duality was also verified on every block for  $m \leq 10$ .)  $\square$

## 5. COLUMN SUBALGEBRAS AND EXPLICIT KERNELS

For  $j \in \mathbb{Z}/3$  put  $c_k^{(j)} = e_{k-1+j}\varphi^k$ . In  $A^s$  one has  $c_a c_b = [3 \mid b] c_{a+b}$ , hence

$$[c_a, c_b] = ([3 \mid a] + [3 \mid b]) c_{a+b}, \quad c_a^{[2]} = [3 \mid a] c_{2a}.$$

These are the root vectors supported in a fixed column of the loop algebra.

**Lemma 5.1.** *The restricted subalgebra  $H^{(j)}$  generated by  $c_1, c_2, c_3$  is  $\text{span}\{c_k : 3 \nmid k\} \oplus \text{span}\{c_{3 \cdot 2^r} : r \geq 0\}$ .*

*Proof.* The right-hand side is closed: a nonzero bracket needs exactly one index divisible by 3, so the sum is not, and a nonzero square needs  $3 \mid a = 3 \cdot 2^r$ . It is generated:  $c_{k+3} = [c_k, c_3]$  for  $3 \nmid k$  and  $c_{3 \cdot 2^{r+1}} = c_{3 \cdot 2^r}^{[2]}$ .  $\square$

Thus in  $L^s/L_{\geq m}^s$  the image of  $H^{(j)}$  has basis  $\{c_k : k \in \mathcal{H}_m\}$ . The squaring chain  $c_3 \rightarrow c_6 \rightarrow c_{12} \rightarrow \dots$  acts as a binary odometer: carries propagate only along the imaginary column vectors of degree  $3 \cdot 2^r$ , while  $c_9, c_{15}, c_{18}, \dots$  are never reached.

**Lemma 5.2.** *Let  $K \subset L^s/L_{\geq m}^s$  be a graded restricted subalgebra with homogeneous basis  $z_1, \dots, z_r$ , and  $\omega_K = z_1 \cdots z_r$ . Then  $\{y \in u_m^s : yz = 0 \ \forall z \in K\} = u_m^s \omega_K$ , of dimension  $2^{N_m - r}$ .*

*Proof.* By PBW, with a complementary basis ordered before the  $z_i$ ,  $u_m^s \cong V \otimes u(K)$  as right  $u(K)$ -modules. So  $y$  annihilates  $K$  iff each coefficient is a right integral of the finite-dimensional Hopf algebra  $u(K)$ ; the integrals form a one-dimensional space [7], spanned by  $\omega_K$  (which spans the top degree of  $u(K)$ ).  $\square$

**Theorem 5.3.** *For every  $m \geq 2$  and  $j \in \mathbb{Z}/3$  let  $\omega_j = \prod_{k \in \mathcal{H}_m} c_k^{(j)}$ . Then*

$$u_m^s \omega_0 + u_m^s \omega_1 + u_m^s \omega_2 \subset \ker R_{\sigma'}.$$

*The  $\omega_j$  are nonzero of degree  $d(m)$  with pairwise distinct  $Q$ -weights. Consequently  $k_m \leq d(m)$  and the kernel of  $R_{\sigma F}$  on  $(u_m)_{d(m)}$  has dimension  $\geq 3$ .*

*Proof.* By Proposition 4.3,  $\sigma' \in c_1 u + c_2 u + c_3 u$  (for  $j = 0$ ; apply  $\rho$  for  $j = 1, 2$ ). If  $y$  annihilates  $c_1, c_2, c_3$  on the right, it annihilates the restricted subalgebra they generate, which is  $H^{(j)}$  by Lemma 5.1, and then  $y\sigma' = 0$ . Lemma 5.2 identifies these  $y$ . The weight of  $c_k^{(0)}$  has  $n_0 - n_2 = 1$  if  $3 \nmid k$  and 0 otherwise, so  $\omega_0$  is not rotation invariant.  $\square$

**Remark 5.4.**  $d(m) = m(m-1)/2$  for  $m \leq 9$ , but  $d(10) = d(9) = 36$ ,  $d(11) = 46$ , and  $d(m) = m^2/3 + O(m \log m)$ ; see Figure 1.

## 6. THE CENTRAL FILTRATION

**Proposition 6.1.**  *$L_m$  is central in  $L/L_{\geq m+1}$  with zero 2-map,  $u_{m+1}$  is a free module over  $\Lambda(L_m)$  with  $u_{m+1}/L_m u_{m+1} = u_m$ , and for every degree (and, in the split model, every  $Q$ -block  $\gamma$ )*

$$\dim \ker R_{\gamma}^{(m+1)} \leq \sum_S \dim \ker R_{\gamma - \text{wt}(S)}^{(m)},$$

*the sum over subsets  $S$  of a homogeneous basis of  $L_m^s$ . Consequently  $k_{m+1} \geq k_m$  and  $\dim \ker R^{(m+1)}/|G_{m+1}| \leq \dim \ker R^{(m)}/|G_m|$ .*

*Proof.*  $[L_i, L_m] \subset L_{\geq m+1}$  and  $L_m^{[2]} \subset L_{2m}$ . With  $J = L_m u_{m+1}$ ,  $J^i/J^{i+1} \cong \Lambda^i(L_m) \otimes u_m$  and  $R_{\sigma F}$  preserves the  $J$ -adic filtration with  $\text{gr } R = 1 \otimes R^{(m)}$ ; apply  $\dim \ker f \leq \dim \ker \text{gr } f$ .  $\square$

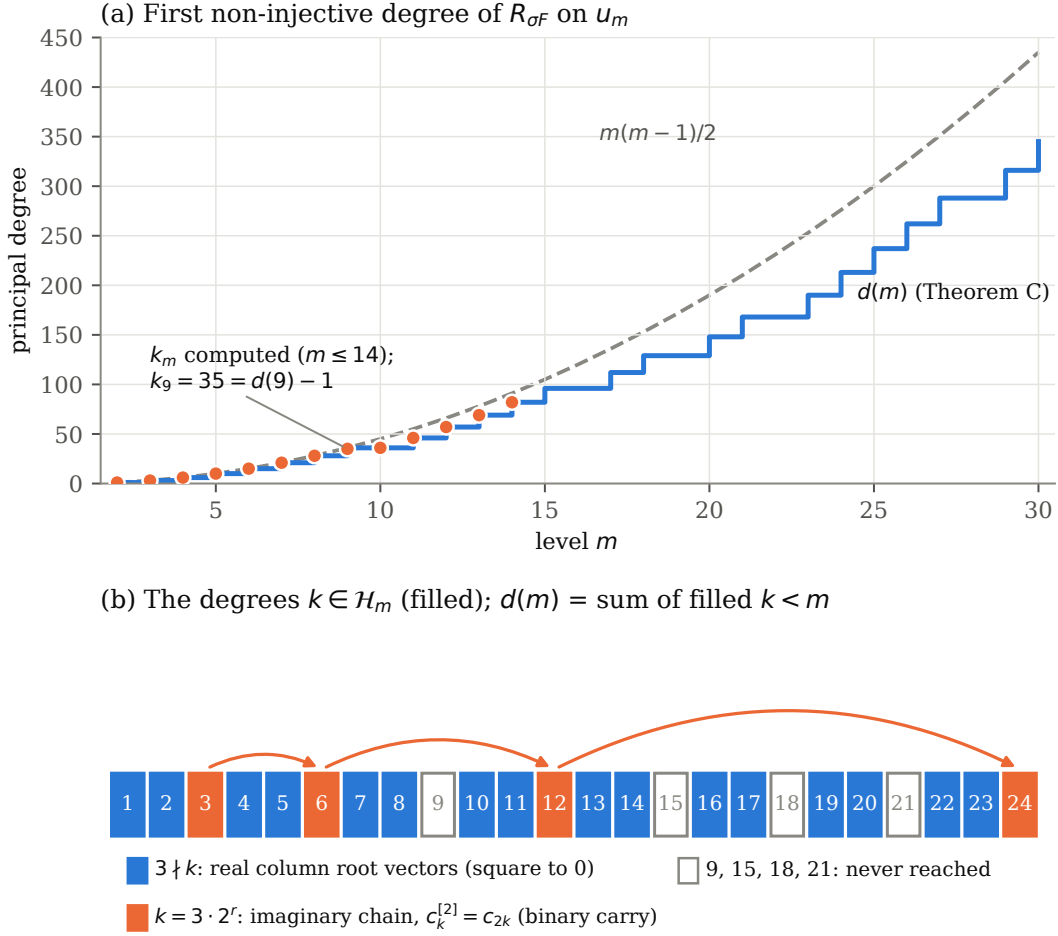


FIGURE 1. (a) The first non-injective degree  $k_m$  of  $R_{\sigma F}$  on  $u_m$  (dots, computed for  $m \leq 14$ ;  $k_{15} \leq 95$ ) against  $d(m)$  of Theorem C (steps) and the naive guess  $m(m-1)/2$  (dashed). (b) The mechanism: the column subalgebra generated by  $c_1, c_2, c_3$  contains  $c_k$  for  $3 \nmid k$  and the imaginary chain  $c_3 \rightarrow c_6 \rightarrow c_{12} \rightarrow c_{24}$  produced by the 2-map (a binary carry), but never  $c_9, c_{15}, c_{18}, c_{21}, \dots$ ; the staircase  $\omega = \prod_{k \in \mathcal{H}_m} c_k$  has degree  $d(m)$  and is annihilated by  $\sigma F$  (Theorem 5.3).

## 7. $\sigma F$ IS A NON-ZERO-DIVISOR

Although  $u(L)$  is not a domain (Theorem 3.1), the leading form of  $F$  is regular.

**Theorem 7.1.** *Left and right multiplication by  $\sigma'$  are injective on  $U = u(\widehat{\mathfrak{n}}) = u(L^s)$ . Consequently  $\sigma F$  is a two-sided non-zero-divisor in  $u(L) = \text{gr } \mathbb{F}_2[[P]]$ ,  $F$  is a two-sided non-zero-divisor in  $\mathbb{F}_2[[P]]$ , and  $k_m \geq m - 3$  for all  $m$ ; in particular  $k_m \geq \max(57, m - 3)$  for  $m \geq 12$ .*

*Proof sketch.* (1) With  $e_i = y_{1,i}$  the three Chevalley generators one checks  $\sigma' = e_0 e_1 e_2 + e_1 e_2 e_0 + e_2 e_0 e_1$ . (2) Let  $\rho^*(x) = (\pi(x) + \text{tr } \pi(x) I)^T$ , where  $\pi$  is the natural representation of Proposition 4.2; this is a graded restricted representation of  $L^s$  over  $\mathbb{F}_2[t]$ , injective on  $L^s$ , with  $e_0 \mapsto tE_{20}$ ,  $e_1 \mapsto E_{01}$ ,  $e_2 \mapsto E_{12}$ , so that  $\rho^*(\sigma') = t \cdot I$ . (3) Let  $\Psi_r$  be the action of  $U$  (via its coproduct) on  $V^*(t_1) \otimes \cdots \otimes V^*(t_r)$ , with independent variables  $t_i$ . Specialising  $t_r = 0$  makes  $\Psi_r(\sigma')$

block upper triangular, and induction gives  $\det \Psi_r(\sigma')|_{(t,0,\dots,0)} = t^{3^r}$ ; hence  $\det \Psi_r(\sigma') \neq 0$  and  $\Psi_r(\sigma')$  is invertible over  $\mathbb{F}_2(t_1, \dots, t_r)$ . (4) The matrix coefficients of  $\rho^*$  generate the graded dual  $U^*$  as an algebra: by graded Nakayama it suffices that they span the indecomposables  $Q(U^*) \cong P(U)^* = (L^s)^*$ , which holds because  $\rho^*$  is injective on  $L^s$ . Thus for every  $y \neq 0$  there is  $r$  with  $\Psi_r(y) \neq 0$ , and  $\Psi_r(y\sigma') = \Psi_r(y)\Psi_r(\sigma') \neq 0$ ; similarly on the left. The statement for  $\sigma F$  follows by Corollary 4.4 (injectivity is invariant under  $\otimes \mathbb{F}_8$ ). Since  $\mathbb{F}_2[[P]]$  is complete for the filtration with  $\text{gr } \mathbb{F}_2[[P]] = u(L)$ , a non-zero-divisor leading form forces  $F$  to be a non-zero-divisor. Finally,  $u_m$  and  $U$  agree in degrees  $< m$ , so  $R_{\sigma'}$  is injective on  $(u_m)_n$  for  $n+3 < m$ , i.e.  $k_m \geq m-3$ . Each step was also checked by machine.  $\square$

**Remark 7.2.** Theorem 7.1 is the “Kaplansky part” of the expected statement  $\text{rk}_P(F) = 1$ ; it does not by itself give  $c_e = 0$ , because  $\mathbb{F}_2[[P]]$  is not known to embed in a division ring. The method gives only a linear bound for  $k_m$ : on  $u_m$  one would need  $\Psi_r(\sigma')^{-1}$  to have monomial denominators, but already  $\det \Psi_3(\sigma') = s_1^{21}(s_1^3 + t_1 t_2 t_3)^2$  with  $s_1 = t_1 + t_2 + t_3$ . A quadratic bound  $k_m \geq m^2/6 - O(m)$ , which would suffice for Theorem 9.4, follows from the statement  $\omega_{K_3} \in u_m \sigma'$ , where  $\omega_{K_3}$  is the product of all letters of degree  $m-3, m-2, m-1$  (by Frobenius duality); we verified it for  $6 \leq m \leq 10$ .

## 8. COMPUTATIONS

| $m$ | $ G_m $  | $\dim C_m$ (exact) | graded cokernel | bound on $c_e$ | $k_m$            |
|-----|----------|--------------------|-----------------|----------------|------------------|
| 2   | $2^3$    | 7                  | 7               | 0.29167        | 1                |
| 3   | $2^6$    | 37                 | 37              | 0.19271        | 3                |
| 4   | $2^8$    | 100                | 100             | 0.13021        | 6                |
| 5   | $2^{11}$ | 547                | 547             | 0.08903        | 10               |
| 6   | $2^{14}$ | 3170               | 3182            | 0.06449        | 15               |
| 7   | $2^{16}$ | <b>10226</b>       | 10226           | 0.05201        | 21               |
| 8   | $2^{19}$ | —                  | <b>64216</b>    | 0.04083        | 28               |
| 9   | $2^{22}$ | —                  | <b>423412</b>   | 0.03365        | <b>35</b>        |
| 10  | $2^{24}$ | —                  | <b>1526176</b>  | <b>0.03032</b> | 36               |
| 11  | $2^{27}$ | —                  | —               | —              | 46               |
| 12  | $2^{30}$ | —                  | —               | —              | <b>57</b>        |
| 13  | $2^{32}$ | —                  | —               | —              | <b>69</b>        |
| 14  | $2^{35}$ | —                  | —               | —              | <b>82</b>        |
| 15  | $2^{38}$ | —                  | —               | —              | $\leq$ <b>95</b> |

TABLE 1. Exact data. Values for  $m \leq 6$  reproduce [1, Table 6.1]; bold entries are new. The bound column is  $\dim C_m/n_m$  for  $m \leq 7$  and (graded cokernel)/ $n_m$  for  $m \geq 8$  (Lemma 2.4).  $k_m$  is the first degree where  $R_{\sigma F}$  fails to be injective on  $u_m$ .

The bounds of the table are plotted in Figure 2.

**Remark 8.1.** The kernel at degree  $k_m$  is 3-dimensional and spanned by  $\omega_0, \omega_1, \omega_2$  for  $m \in \{2, \dots, 8\} \cup \{10, \dots, 14\}$ . The case  $m = 9$  is exceptional:  $k_9 = 35 = d(9) - 1$ , and the kernel there is spanned by the rotations of a “lowered staircase”  $\eta$ , which is not annihilated by any single letter and arises from a cancellation inside  $\eta\sigma'$ ; it leaves the kernel at  $m = 10$ . In particular the empirical rule  $k_m = m(m-1)/2$ , valid for  $m \leq 8$ , fails for  $m \geq 9$ .

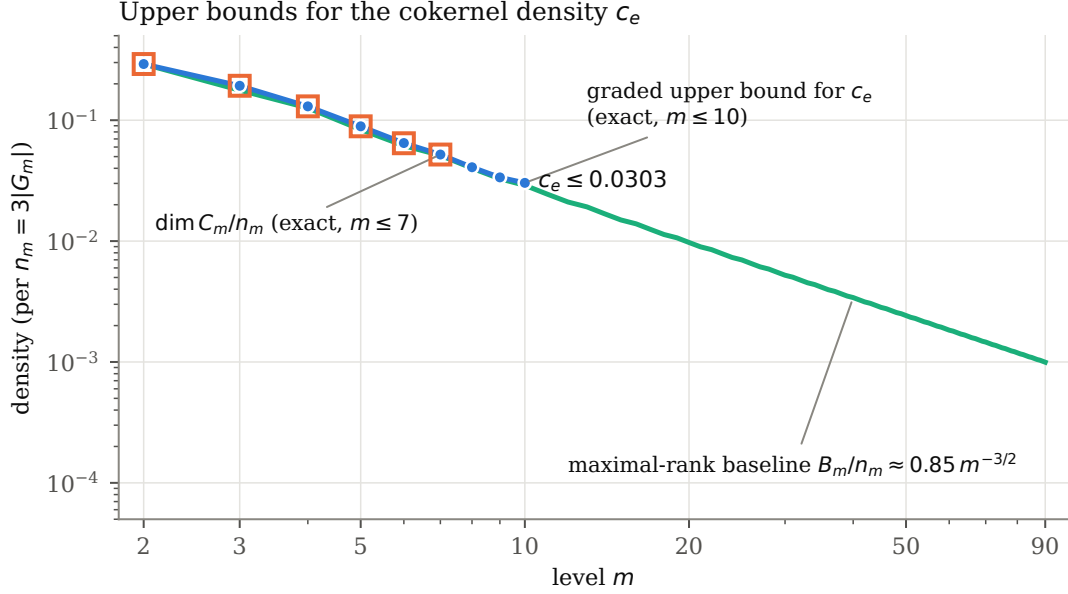


FIGURE 2. Upper bounds for  $c_e$  (log-log). Squares: exact  $\dim C_m/n_m$  ( $m \leq 7$ ). Dots: the graded upper bound  $\dim \text{coker}(R_{\sigma F}|u_m)/n_m$  (Lemma 2.4), exact for  $m \leq 10$ , giving  $c_e \leq 0.0303$ . Line: the maximal-rank baseline  $B_m/n_m$  (Section 9), computed exactly for  $m \leq 90$ ; the graded bound equals the baseline plus the excess  $e_m/3$ , and  $c_e = 0$  is equivalent (at the graded level) to  $e_m \rightarrow 0$ .

### 9. MAXIMAL-RANK BASELINE AND EXCESS

Throughout,  $h_n = \dim(u_m)_n$ ,  $\text{top}$  is the top degree of  $u_m$ ,  $\text{mid} = \text{top}/2$ , and  $\ker_n = \dim \ker(R_{\sigma}|(u_m)_n)$ . Let  $X_m$  be the degree of a uniformly random PBW monomial of  $u_m$ , i.e.  $X_m = \sum_{k < m} \sum_{i=1}^{d_k} k \varepsilon_{k,i}$  with independent fair bits  $\varepsilon_{k,i}$ , so that  $h_n = |G_m| \Pr[X_m = n]$  and  $\mathbb{E}X_m = \text{mid}$ .

Trivially  $\dim \ker R_{\gamma} \geq \max(0, \dim u_{\gamma} - \dim u_{\gamma+\delta})$ . Put  $B_m^Q = \sum_{\gamma} \max(0, \dim u_{\gamma} - \dim u_{\gamma+\delta})$ , and define the excess  $e_m$  by

$$x_m := \frac{\dim \ker R_{\sigma F}|u_m}{|G_m|} = \frac{B_m^Q}{|G_m|} + e_m, \quad c_e \leq x_m/3.$$

**Lemma 9.1.** *There is an absolute constant  $A$  with  $\max_n \Pr[X_m = n] \leq A m^{-1/2}$  and  $B_m := \sum_n \max(0, h_n - h_{n+3}) \leq A m^{-1/2} |G_m|$ .*

*Proof.* For each  $k < m - 3$  with  $k \equiv 1, 2 \pmod{6}$  pick one letter of degree  $k$  and one of degree  $k + 3$ ; these  $M \geq m/3 - 2$  pairs are disjoint. Let  $T$  be the number of pairs in which exactly one of the two letters occurs. Conditionally on  $T$  and on all other bits,  $X_m = Y + 3 \text{Bin}(T, \frac{1}{2})$  with  $Y$  fixed. Hence  $\max_n \Pr[X_m = n]$  and the total variation distance  $d_{TV}(X_m, X_m + 3)$  are bounded by  $\mathbb{E}[\min(1, A_0 T^{-1/2})]$ , and  $T \sim \text{Bin}(M, \frac{1}{2})$  gives  $O(M^{-1/2})$ . Finally  $\sum_n (h_n - h_{n+3}) = 0$ , so  $B_m = \frac{1}{2} \sum_n |h_n - h_{n+3}| = |G_m| d_{TV}(X_m, X_m + 3)$ .  $\square$

Since  $B_m^Q$  and  $B_m$  agree numerically to four digits, Lemma 9.1 shows rigorously that the degree-wise baseline is  $O(m^{-1/2})$ ; the exact values for  $m \leq 90$  follow  $B_m/|G_m| \approx 2.55 m^{-3/2}$ , as

a local central limit theorem predicts ( $\text{Var } X_m = \frac{1}{4} \sum_{k < m} k^2 d_k \sim 2m^3/9$ ). The excess is small:  $e_m = 0.0025, 0.0022, 0.0027, 0.0041$  for  $m = 7, \dots, 10$ , symmetric about the middle degree.

**Conjecture 9.2.**  $e_m \rightarrow 0$ . *Equivalently (given the baseline asymptotics) the graded cokernel density tends to 0, and hence  $c_e = 0$ .*

Conjecture 9.2 is a Lefschetz-type maximal-rank statement for multiplication by the weight- $\delta$  element  $\sigma'$  on the truncated restricted enveloping algebras of  $\widehat{\mathfrak{n}}$ ; compare [9] for the commutative theory. The exceptional element  $\eta$  at  $m = 9$  shows that a proof cannot rest on annihilators alone.

**9.1. The excess is a special-position effect.** The weight- $\delta$  subspace of  $u_m^s$  is 6-dimensional (spanned by  $y_{3,0}, y_{3,1}, y_{2,q}y_{1,q+1}$  and  $y_{1,0}y_{1,1}y_{1,2}$ ), so all 63 nonzero  $\mathbb{F}_2$ -elements can be tested.

**Proposition 9.3** (proof by computation). *For  $4 \leq m \leq 8$ , a generic element of  $(u_m^s)_\delta$  (over  $\overline{\mathbb{F}_2}$ ) has maximal rank on every  $Q$ -block, i.e. excess 0. For  $m = 9$  its excess is at most 18, concentrated in 18 small square blocks near the middle degree. By contrast  $\sigma'$  has excess 3, 30, 156, 162, 1136, 11200 for  $m = 4, \dots, 9$ .*

(Maximal rank is certified blockwise by exhibiting one element, over  $\mathbb{F}_2, \mathbb{F}_{16}$  or  $\mathbb{F}_{256}$ , of maximal rank; rank is lower semicontinuous.) Thus the truncated algebras  $u_m^s$  have the maximal-rank property for weight  $\delta$  almost perfectly, and the excess of  $\sigma'$  comes from its special position. Semicontinuity bounds  $\text{rk } R_{\sigma'}$  in the wrong direction, so this does not by itself prove Conjecture 9.2.

## 9.2. A conditional proof.

**Theorem 9.4.** *Suppose that for some constant  $C$  and infinitely many  $m$ :*

- (H1)  $\ker_n \leq C h_{n-k_m}$  for  $k_m \leq n \leq \text{mid}$ , and
- (H2)  $k_m \geq d(m) - 1$ .

*Then along these  $m$*

$$x_m \leq 2C \Pr[X_m \leq \text{mid} - k_m] + 6 \max_n \Pr[X_m = n] \leq 2C e^{-m/4+O(1)} + 6A m^{-1/2},$$

*and consequently  $c_e = 0$ . In (H2) it suffices that  $k_m/m^{3/2} \rightarrow \infty$ .*

*Proof.* Let  $r_n$  be the rank of  $R_{\sigma'} : (u_m)_n \rightarrow (u_m)_{n+3}$ , so  $\ker_n = h_n - r_n$ . By the duality of Corollary 4.4 (summed over blocks of a fixed principal degree),  $\ker_n = h_{\text{top}-n} - r_{\text{top}-n-3}$ . Writing  $j = \text{top} - n - 3$  this reads  $\ker_n = (h_{j+3} - h_j) + \ker_j$ . Summing over  $n > \text{mid}$ , the terms  $h_{j+3} - h_j$  telescope to at most  $3 \max_n h_n$ , and the degrees  $n$  within 3 of  $\text{mid}$  contribute at most  $3 \max_n h_n$  more; hence

$$\dim \ker R_{\sigma'} \leq 2 \sum_{n \leq \text{mid}} \ker_n + 6 \max_n h_n.$$

By (H1),  $\sum_{n \leq \text{mid}} \ker_n \leq C \sum_{j \leq \text{mid}-k_m} h_j = C |G_m| \Pr[X_m \leq \text{mid} - k_m]$ . Hoeffding's inequality for  $X_m$  (a sum of independent variables with ranges  $k$ ) gives  $\Pr[X_m \leq \text{mid} - k_m] \leq \exp(-2k_m^2 / \sum_{k < m} k^2 d_k)$ ; with  $k_m \geq d(m) - 1 \sim m^2/3$  and  $\sum_{k < m} k^2 d_k \sim 8m^3/9$  the exponent is  $-m/4 + O(1)$ , and it tends to  $-\infty$  as soon as  $k_m/m^{3/2} \rightarrow \infty$ . The last term is bounded by Lemma 9.1. Since  $x_m$  is non-increasing (Proposition 6.1), a subsequence suffices, and  $c_e \leq x_m/3$ .  $\square$

Hypothesis (H1) holds with  $C = 3$  for  $5 \leq m \leq 10$  (the ratio  $\ker_n / h_{n-k_m}$  equals 3 at  $n = k_m$  and decreases afterwards), and (H2) holds for  $m \leq 14$ : an exact layer-wise lifting from level 10 gives  $k_{12} = 57 = d(12)$ ,  $k_{13} = 69 = d(13)$  and  $k_{14} = 82 = d(14)$  (all  $\rho$ -orbits of blocks were computed; for  $m = 14$  the largest block has dimension 16 769 439), and in each case the kernel at  $d(m)$  is exactly the span of  $\omega_0, \omega_1, \omega_2$ . Hence  $R_{\sigma F}$  is injective in all degrees  $\leq 81$  for every

$m \geq 14$  (Proposition 6.1). For  $m = 15$  we found the predicted exceptional element: an element  $\eta_{15} \equiv y_{13,1} \omega_0^{(14)} \pmod{L_{14}}$  of degree  $95 = d(15) - 1$  with  $\eta_{15} \sigma' = 0$ , so  $k_{15} \leq 95$ ; like  $\eta$  at  $m = 9$ , it is not annihilated by any single letter and it leaves the kernel at  $m = 16$ . Injectivity below 95 was checked only on the smaller blocks of degrees 94 and 95 (the only lower bound proved is  $k_{15} \geq k_{14} = 82$ ).

**Remark 9.5.** The condition  $k_m/m^{3/2} \rightarrow \infty$  is sharp for this method: if  $k_m = O(m^{3/2})$ , the central limit theorem keeps  $\Pr[X_m \leq \text{mid} - k_m]$  bounded below. One can also show that  $k_m \rightarrow \infty$  if and only if  $R_{\sigma'}$  is injective on the untruncated algebra  $u(\widehat{\mathfrak{n}})$ , in which case  $k_m \geq m - 3$ ; otherwise  $k_m$  is eventually constant ( $\geq 57$ ). Moreover, the column annihilators of Section 5 cannot by themselves prove (H2): from  $m = 7$  on, the dual witnesses for the disappearance of  $\omega_j$  at a jump step are of cancellation type.

For  $m = 11$  a partial computation (all blocks of dimension  $\leq 2 \cdot 10^4$ , remaining blocks bounded via Proposition 6.1) gives  $0.00051 \leq e_{11} \leq 0.01693$ .

## 10. GRADED ZERO DIVISORS DO NOT LIFT

**Proposition 10.1** (proof by computation). (1) *For every  $0 \neq v \in V$  and every  $x \in \mathbb{F}_2[[P]]$  with leading form  $v$ ,  $x^2 \in I^9 \setminus I^{10}$ . The obstruction is a nonzero class in  $\ker \text{ad}_v / \text{im ad}_v$  in degree 9.*

(2) *For every pair  $(v, w) \in (V \setminus 0)^2$  and all lifts  $x, y$ ,  $xy \notin I^{11}$ .*

(3) *For the (3, 14) family of the Remark in Section 3,  $xy \in I^{18} \setminus I^{19}$  for all lifts.*

These statements are exact: they solve the lifting equations as linear systems in  $\text{gr } \mathbb{F}_2[[P]]$ , enumerating the full affine solution space where the equations become nonlinear (e.g. all  $2^9$  choices in (2)). Moreover, for the lifts  $\tilde{v}$  the kernel density of right multiplication on  $\mathbb{F}_2[G_m]$  is 0.375, 0.279, 0.234 ( $m = 5, 6, 7$ ), decreasing like that of a generic element of  $I^4$  (0.332, 0.244, 0.203), whereas in the graded algebra it stays at 0.625. This is consistent with the pro-2 Kaplansky conjecture for  $P$ , but of course does not prove it.

## 11. QUESTIONS

**Question 11.1.** Prove Conjecture 9.2, or directly that  $\text{rk}_P(F) = 1$ . By Theorem 9.4 it suffices to prove (H1) and (H2): the kernel of  $R_{\sigma'}$  below the middle degree grows no faster than a fixed multiple of the free module generated in degree  $k_m$ , and the threshold stays within one of  $d(m)$ .

**Question 11.2.** Is  $k_m \in \{d(m) - 1, d(m)\}$  for all  $m$ ? More precisely, is it true that  $k_m = d(m) - 1$  if  $3 \mid m$ ,  $m \geq 9$  and  $m \neq 3 \cdot 2^r$ , and otherwise  $k_m = d(m)$  with kernel spanned by  $\omega_0, \omega_1, \omega_2$ ? This was formulated from the data for  $m \leq 12$  and then confirmed for  $m = 13, 14$  ( $k_{13} = 69$ ,  $k_{14} = 82$ ); for  $m = 15$  the predicted exceptional kernel element in degree 95 exists.

**Question 11.3.** Is there a filtration of  $\mathbb{F}_2[[P]]$  (for instance a weight filtration adapted to the twisted form of  $\widehat{\mathfrak{n}}$ ) whose associated graded ring is a domain? By Proposition 2.2 such a ring would have subexponential growth, hence be Ore; if moreover  $\mathbb{F}_2[[P]]$  then embeds continuously into a division ring  $Q$  with a discrete valuation, [4, Thm. 7.1] ( $\varphi^\# \text{rk}_Q \leq \text{rk}_P$ ) would give  $\text{rk}_P(x) = 1$  for all  $x \neq 0$ , in particular  $c_e = 0$ .

**Question 11.4.** The same construction applies to other cyclic division algebras  $(\mathbb{F}_{q^3}(t)/\mathbb{F}_q(t), \sigma, t)$  and to higher degree. Do the analogues of Theorems B and C hold, with  $A_{d-1}^{(1)}$  in place of  $A_2^{(1)}$ ?

## APPENDIX A. COMPUTATIONAL METHODS

All code, data and logs are in the repository <https://github.com/Ruqing1963/ancfft-ce-density> (directories `code/`, `data/`, `results/`); the figures are produced by `code/make_data_and_figures.py`. The group  $G_m$  is realised as normalised truncated skew power series in  $\mathbb{F}_8[[\varphi; \sigma]]/(\varphi^m)$ , with conventions ported line by line from Chen's published scripts [1]. Ranks over  $\mathbb{F}_2$  use a bit-packed elimination with 64-column panels and Four-Russians tables (numba). Independent checks: (a)  $\dim C_m$  for  $m \leq 6$  reproduces [1]; (b)  $\dim C_7$  was computed from three matrices of provably equal rank (transpose, left multiplication, random row and column permutations); (c)  $u(L)$  was implemented twice (the  $\mathbb{F}_8$ -model and the split model, with independent straightening codes) and checked against Jennings products in  $\mathbb{F}_2[G_N]$  for  $N \leq 12$ ; (d) the zero divisor of Theorem 3.1 was confirmed by plain group multiplication in  $\mathbb{F}_2[G_9]$ ; (e) the graded cokernels for  $m = 9, 10$  use the block decomposition of Corollary 4.4; for  $m = 10$  one block per rotation orbit was computed and the remaining blocks were spot-checked.

**Acknowledgements.** Computations and parts of the analysis were carried out with the assistance of an AI system (Claude, Anthropic).

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